

(TR-104) PYTHON WITH AI/ML

Training Day 82 Report

Date: 5 May 2026

Topic Covered: RAG Revision, LangChain Fundamentals & AI Concepts Review

The eighty-second day of training focused on revising previously covered concepts related to Retrieval-Augmented Generation (RAG), LangChain, document retrieval workflows, and AI agent architectures to strengthen foundational understanding.

- Revised the core concepts of Retrieval-Augmented Generation (RAG) and its role in improving LLM responses.
- Reviewed document ingestion, chunking strategies, embeddings, and vector database fundamentals.
- Studied retriever mechanisms and context retrieval workflows used in RAG pipelines.
- Revised LangChain fundamentals, including chains, prompts, tools, agents, and output parsers.
- Reviewed the interaction between Large Language Models and external knowledge sources through LangChain.
- Strengthened understanding of tool-based agent architectures and query-processing workflows.
- Revised prompt engineering concepts for generating accurate and context-aware responses.
- Reviewed the integration of RAG pipelines with LLMs for efficient information retrieval and response generation.
- Revisited previously implemented project concepts and analyzed their practical applications.

Overall, the training enhanced understanding of RAG architectures, LangChain components, retrieval workflows, prompt engineering, and AI-powered application development.